Computer Systems

13th November 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

		Marks
Part 1: Computer Systems and Hardware		
1a)	Describe TWO (2) main types of computer systems.	**4**
	Mark Scheme: ***1 mark for type and 1 mark for correct associated description.*** ***Any two from the following:*** ***Mainframe computers (1 mark) - Powerful, centralised systems used for large-scale computing tasks (1 mark).*** ***Personal computers (1 mark) - General-purpose systems for everyday tasks (1 mark).*** ***Embedded systems (1 mark) - Systems embedded in devices for specific tasks (1 mark).***	
1b)	Explain the fundamental function of hardware **and** software in a computer system.	**2**
	Mark Scheme: ***1 mark for hardware role and 1 mark for software role.*** ***Hardware provides the physical components (e.g., CPU, memory, storage) (1 mark) while software instructs the hardware on how to perform tasks (1 mark).***	
1c)	Explain the role of the operating system **and** its role in interfacing with hardware and software. Provide ONE (1) example of how the operating system works in this relationship. ***1 mark for hardware-software interaction concept and 1 mark for example.***	**2**
	The operating system serves as an intermediary, allowing software applications to interact with the hardware effectively (1 mark). ***For example, when a program is run, the software instructions are translated by the CPU to perform computations or I/O operations (1 mark).*** ***Answers may differ from the above suggestions. There may be alternative suitable answers provided by students.***	

© NCC Education Limited 2024

2a)	Describe the function of THREE (3) modern computer hardware components – the CPU, Memory and I/O devices.	**3**
	Mark Scheme: ***1 mark per correct description*** ***CPU - Executes program instructions and performs calculations. (1 mark)*** ***Memory - Stores data and instructions temporarily or permanently, depending on the type (e.g., RAM for temporary storage, hard drives for permanent storage). (1 mark)*** ***Input/Output devices: Facilitate interaction between the user (peripherals) and the computer (e.g., keyboard for input, monitor for output). (1 mark)***	
2b)	Explain the function of the ALU **and** control unit in the CPU.	**2**
	Mark Scheme: ***ALU (Arithmetic Logic Unit): Performs arithmetic and logical operations on data, such as addition, subtraction and comparison. (1 mark)*** ***Control Unit: Directs the operations of the CPU by fetching, decoding and executing instructions in sequence, managing the flow of data between the CPU and memory. (1 mark)***	
3a)	Identify FOUR (4) features of a RISC that allow it to operate simply and efficiently.	**4**
	Mark Scheme: ***<u>Any 4 from below to a maximum of 4 marks:</u>*** ***Large set of registers (32, 64, 128 or more) (1 mark)*** ***Uses registers for passing parameters to functions (1 mark)*** ***Uses registers to return values from functions (1 mark)*** ***Global variables stored in registers (1 mark)*** ***Low number of instructions (less than 100) (1 mark)*** ***Reduced number of addressing modes (1 mark)*** ***Fixed size instructions (64 / 128 bytes) (1 mark)***	
3b)	Briefly describe ONE (1) advantage of a multi-core **and** ONE (1) advantage of a quantum computing systems. State which of these systems might be better to use for potentially breaking traditional encryption algorithms.	**3**

© NCC Education Limited 2024

	Mark Scheme: ***Multi-core Systems: Parallel processing OR Energy Efficiency OR Improved Multi-tasking (1 mark).*** ***Quantum computing Systems: Exponential Processing Power OR Solving Complex Problem (1 mark).*** ***Quantum computing is best utilised for breaking encryption algorithms (1 mark).***	
		Total 20 Marks

© NCC Education Limited 2024

Part 2: Operating Systems		
4a)	State the FOUR (4) main components of an operating system.	**4**
	Mark Scheme: ***Kernel (1 mark), File System (1 mark), User Interface (1 mark), Device Drivers (1 mark).***	
4b)	State the purpose of timesharing and explain how it works as part of a virtual machine. Describe ONE advantage and ONE disadvantage of timesharing.	**4**
	Mark Scheme: ***Timesharing is a system that allows multiple users or processes to share computing resources (1 mark) by allocating a specific amount of time for each task in a rotating sequence (1 mark).*** ***Stated Advantage and Disadvantage must be inherently <u>different from one another</u>.*** ***Advantages (Only 1 from):*** • ***Resource Efficiency: Timesharing maximizes the use of CPU and memory resources. (1 mark)*** • ***Perceived Fairness: A timeshared system works by allocating each user a measurable time slot that is equal for every user. (1 mark)*** • ***Cost-Effective: Timesharing systems can support multiple users on a single machine. (1 mark)*** • ***Supports Multitasking: Allows multiple applications to run concurrently. (1 mark)*** • ***Flexibility: Timesharing systems can adjust the allocation of resources dynamically, helping to balance loads based on demand. (1 mark)*** ***Disadvantages (Only 1 from):*** • ***Reduced Performance Under High Load: As more users or processes are added, each one gets a smaller slice of time, which can reduce performance. (1 mark)*** • ***Complexity of Management: Requires sophisticated scheduling and resource management to ensure fair distribution and prevent issues, such as deadlock. (1 mark)*** • ***Security and Privacy Risks: Security concerns if isolation between user sessions is not properly maintained. (1 mark)*** • ***Dependency on Stable Software: Timesharing requires reliable system software to handle the frequent switching of processes. (1 mark)***	

© NCC Education Limited 2024

5a)	State **and** describe TWO (2) examples of operating system resource management.	**4**
	Mark Scheme: ***Any <u>two</u> from below. One mark for OS resource management and 1 mark for a correct associated description.*** ***1. CPU Scheduling (1 mark): The OS manages which process gets CPU time and for how long, optimising efficiency and responsiveness (1 mark).*** ***2. Memory Management (1 mark): The OS controls how memory is allocated to processes and ensures memory is freed when no longer needed, including virtual memory management (1 mark).*** ***3. Device Management (1 mark): The OS controls and coordinates communication between applications and peripheral devices like printers, keyboards and storage drives, using device drivers to ensure smooth and reliable I/O operation (1 mark).*** ***4. File System Management (1 mark): The OS organises, stores and retrieves files on storage devices, providing a hierarchical structure for easy navigation and ensuring that data is accessible and secure through permissions and file locks (1 mark).*** ***5. Network Management (1 mark): The OS manages the system's networking functionality by controlling how data is sent and received over a network, including managing bandwidth and IP addresses, ensuring reliable communication between devices (1 mark).***	
5b)	Explain the interaction between an operating system and hardware.	**2**
	Mark Scheme: ***The OS serves as an intermediary between software and hardware (1).*** ***The OS ensures programs can communicate with hardware devices without needing direct access. (1 mark)***	
5c)	Outline FOUR (4) distinct characteristics of a RTOS (Real Time Operating System)'s operation that you may find in an embedded system such as a motorbike's electronic control unit (ECU).	**4**

© NCC Education Limited 2024

	Mark Scheme: ***<u>Any 4 from below to a maximum of 4 marks:</u>*** ***Task scheduling - mention of pre-emptive OR priority-based scheduling (1 mark)*** ***Fixed memory - dynamic memory OR static memory (1 mark)*** ***Deterministic timings – mention of interrupts OR delays OR control OR context switching OR task creation (1 mark)*** ***Task synchronisation (1 mark)*** ***Reliability and Fault Tolerance – mention of mechanisms for task monitoring OR fault recovery OR system diagnostics (1 mark)*** ***Interrupt Handling – mention of fast/efficient handling of real-time interrupts (1 mark)***	
5d)	Explain the role of virtual memory in an operating system **and** discuss how it improves system performance.	**2**
	Mark Scheme: ***Correctly identifying virtual memory as a feature of memory management that extends physical RAM using disk space (1 mark).*** ***Correct explanation of how it improves performance, such as allowing more applications to run simultaneously by providing additional memory or preventing system crashes due to insufficient RAM (1 mark).***	
		Total 20 Marks

© NCC Education Limited 2024

Part 3: Numbers and Logic

6a) Convert the following decimal number to binary: **45**
Show your working. **2**

Mark Scheme:

101101 (1 mark)
Workings demonstrating value of each bit and/or calculation of individual values for total (1 mark)

6b) Perform the following binary addition: 1010 + 0111 **3**

Mark Scheme:
1 mark for attempting addition but partial results.
3 marks for correct answer and evidence of carry over.

1010 (~10)
+ ***0111 (~7)***
010001 **(~17) accept 10001**

6c) Explain what is meant by binary coded decimal (BCD) **and** explain how it differs from binary. **4**

Mark Scheme:

BCD is a representation of decimal digits using binary (1 mark).

Each decimal digit (0–9) is represented by a 4-bit binary number (1 mark).

Explaining that BCD allows decimal numbers to be stored directly in binary form (1 mark).

Explaining how BCD differs from regular binary (i.e., binary represents the entire number in one sequence, whereas BCD represents each decimal digit individually) (1 mark).

7a) Complete the following truth table for the logic function: A AND (B OR C). **9**

A	***B***	***C***		***A AND (B OR C)***
0	*0*	*0*	*0*	
0	*0*	*1*		
0	*1*	*0*	*1*	
0	*1*	*1*		*0*
1	*0*	*0*		*0*
1	*0*	*1*	*1*	*1*
1	*1*	*0*		
1	*1*	*1*	*1*	*1*

© NCC Education Limited 2024

Mark Scheme: ***1 mark for each of the missing table elements.***

A	*B*	*C*	*B OR C (1 mark)*	*A AND (B OR C)*
0	*0*	*0*	*0*	*0* ***(1 mark)***
0	*0*	*1*	*1* ***(1 mark)***	*0* ***(1 mark)***
0	*1*	*0*	*1*	*0* ***(1 mark)***
0	*1*	*1*	*1* ***(1 mark)***	*0*
1	*0*	*0*	*0* ***(1 mark)***	*0*
1	*0*	*1*	*1*	*1*
1	*1*	*0*	*1* ***(1 mark)***	*1* ***(1 mark)***
1	*1*	*1*	*1*	*1*

7b) Convert the hexadecimal number B37 to its decimal value. Show your working. **2**

Mark Scheme:

$(B37)_{16} = (11 \times 16^2) + (3 \times 16^1) + (7 \times 16^0) = (2871)_{10}$

$= 2816 + 8 + 7 = 2871$

1 mark for correct answer.
1 mark for working.

Total 20 Marks

© NCC Education Limited 2024

Part 4: Software Hierarchy and Computer Programs		
8a)	Define the term "*addressing mode*" **and** state TWO (2) examples of addressing modes.	**4**
	Mark Scheme ***An addressing mode refers to the way in which the operand of an instruction is specified in machine code (1 mark).*** ***It determines how the CPU locates the data that an instruction needs to process (1 mark).*** ***Immediate addressing mode (1 mark) and Direct addressing mode (1 mark).***	
8b)	Explain the fetch-decode-execute cycle of instruction processing.	**4**
	Mark Scheme ***1 mark for purpose. 1 mark for each correct description of the fetch-decode-execute cycle.*** ***The fetch-decode-execute cycle is the process through which a CPU processes an operation (1 mark).*** ***Fetch: Loading of the instruction identified by The Program Counter (PC) into the IR (1 mark)*** ***Decode: The CPU decodes the instruction in the IR to determine what operation is to be performed and identifies the operands (data) required for that operation. (1 mark)*** ***Execute: This can involve performing arithmetic or logical operations, moving data from one location to another or interacting with input/output devices. (1 mark)***	
8c)	Describe an interrupt **and** explain how an interrupt mechanism works in computer systems.	**2**

© NCC Education Limited 2024

	Mark Scheme ***An interrupt is a signal sent to the CPU that indicates an event needing immediate attention (1 mark).*** ***It temporarily halts the current execution of a program, allowing the CPU to respond to important events, such as input from peripherals or hardware failures (1 mark).*** ***Answers may differ from the above suggestions. There may be alternative suitable answers provided by students.***	
9a)	Compare the roles of compilers **and** interpreters in software development.	**6**
	Mark Scheme ***Compiler Explanation:*** ***A compiler is a program that translates high-level programming code into machine code (1 mark)*** ***The compiler processes the entire source code at once and generates an executable file (1 mark).*** ***Interpreter Explanation:*** ***An interpreter is a program that translates high-level programming code into machine code line-by-line or statement-by-statement at runtime (1 mark).*** ***The interpreter does not produce an intermediate executable file; instead, it executes the source code directly (1 mark).*** ***Comparison: Up to 2 marks:*** ***<u>Execution Method:</u>*** ***Compiler: Translates the entire code before execution, resulting in a standalone executable file. (1 mark)*** ***Interpreter: Translates and executes the code line-by-line or statement-by-statement at runtime. (1 mark)*** ***<u>Speed of Execution:</u>*** ***Compiler: Generally faster execution, as the code is precompiled into machine code. (1 mark)*** ***Interpreter: Slower execution, as it interprets the code at runtime. (1 mark)*** ***<u>Error Detection:</u>*** ***Compiler: Detects errors at compile time, meaning that the entire program must be error-free before execution. (1 mark)*** ***Interpreter: Detects errors during execution, allowing developers to address issues as they arise. (1 mark)*** ***Answers may differ from the above suggestions. There may be alternative suitable answers provided by students.***	

© NCC Education Limited 2024

9b)	Micro-controllers are widely used in several different application areas. For example, the automotive industry. Provide FOUR (4) additional examples of application areas where micro-controllers are used.	**4**
	Mark Scheme: ***<u>Any 4 from below to a maximum of 4 marks:</u>*** ***Security industry (1 mark)*** ***Domestic appliances (1 mark)*** ***Military equipment (1 mark)*** ***Scientific instruments (1 mark)*** ***Computer peripherals (1 mark)*** ***Entertainment industry (1 mark)*** ***Correct answers are not limited to the above. There may be alternative suitable answers provided by students.***	
		Total 20 Marks

© NCC Education Limited 2024

Part 5: Multiple Choice Questions (20 Marks)			
10a)	Which of the following best describes the function of an operating system.		**2**
	i. To perform calculations and execute programs	ii. To manage computer hardware and software resources	
	iii. To provide network connectivity	iv. To compile high-level programming languages	
	Mark Scheme: ***ii) To manage computer hardware and software resources (2 marks)***		
10b)	What is the primary purpose of memory in a computer system?		**2**
	i. To store the operating system only	ii. To execute arithmetic operations	
	iii. To temporarily hold data and instructions for quick access	iv. To provide permanent data storage	
	Mark Scheme: ***iii) To temporarily hold data and instructions for quick access (2 marks)***		
10c)	Which component of the computer system is primarily responsible for temporarily holding data and instructions for quick access during processing?		**2**
	i. Control Unit	ii. Memory Unit	
	iii. ALU	iv. Cache	
	Mark Scheme: ***iv) Cache (2 marks)***		
10d)	In data communications, which of the following is **NOT** considered a goal?		**2**
	i. Reliability	ii. Security	
	iii. Transmission efficiency	iv. Cost-effectiveness	
	Mark Scheme: ***iv) Cost-effectiveness (2 marks)***		
10e)	What is the binary representation of the hexadecimal value FC. Select ONE (1) option.		**2**
	i. 11111100	ii. 10100011	
	iii. 00110101	iv. 11001111	
	Mark Scheme: ***i) 11111100 (2 marks)***		

© NCC Education Limited 2024

11a)	Which network topology has all nodes connected to a central hub?		**2**
	i. Bus Topology	ii. Ring Topology	
	iii. Mesh Topology	iv. Star Topology	
	Mark Scheme: ***iv) Star Topology (2 marks)***		
11b)	Which of the following is a characteristic of quantum computing?		**2**
	i. Uses classical bits for processing	ii. Utilises qubits to represent and process information	
	iii. Operates on a binary logic framework	iv. Relies solely on traditional hardware components	
	Mark Scheme: ***ii) Utilises qubits to represent and process information (2 marks)***		
11c)	Which of the following types of software manages the hardware of a computer?		**2**
	i. Application software	ii. System software	
	iii. Middleware	iv. Firmware	
	Mark Scheme: ***ii) System software (2 marks)***		
11d)	What is the primary purpose of the Hypertext Transfer Protocol?		**2**
	i. To send email messages between servers	ii. To establish a secure connection between two devices	
	iii. To facilitate the transfer of web pages and resources over the Internet	iv. To manage network traffic and routing	
	Mark Scheme: ***iii) To facilitate the transfer of web pages and resources over the Internet (2 marks)***		
11e)	Which device is used to connect multiple computers within a local area network (LAN) and manage data traffic between them?		**2**
	i. Modem	ii. Router	
	iii. Switch	iv. Hub	
	Mark Scheme: ***iii) Switch (2 marks)***		
			Total 20 Marks

End of paper

© NCC Education Limited 2024

Learning Outcomes Matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1,2,3	1,6	Yes
4,5	2	Yes
6,7	3	Yes
8,9	4,5	Yes
10, 11	1,2,6,7	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand and identify the main types and components of computer systems	None or inadequate description of computer systems and components	Some credible but incomplete attempt to demonstrate understanding of computer systems and components	Limited but satisfactory demonstration of understanding of computer systems and components	Good demonstration of understanding of computer systems and components	Excellent demonstration of understanding of computer systems and components
Describe the structure and role of modern operating systems	None or very basic description of components of operating systems	A very limited and basic description of components of operating systems	Limited and above basic description of components of operating systems	Good description of most components of operating systems	Excellent and full description of all components of operating systems
Understand and work with binary numbers and computer logic	No demonstration of working with binary numbers and logic operators	A basic demonstration of restricted range of working with binary numbers and logic operators	An above basic demonstration of working with restricted range of binary numbers and logic operators	A good demonstration of working with a wide range of binary numbers and logic operators	An excellent demonstration of working with full range of binary numbers and logic operators
Understand the essential structure of computer programs	None or trivial explanation of instruction sets, addressing modes and interrupts	Limited understanding and explanation of instruction sets, addressing modes and interrupts	Basic understanding and explanation of instruction sets, addressing modes and interrupts	Good understanding and explanation of instruction sets, addressing modes and interrupts	Excellent understanding and explanation of instruction sets, addressing modes and interrupts
Explain the hierarchy of computer software	No or wrong explanation of computer software hierarchy	Limited and very basic explanation of computer software hierarchy	Basic but limited explanation of computer software hierarchy	Good and competent explanation of computer software hierarchy	Excellent and complete explanation of computer software hierarchy
Describe alternative computer	No or insufficient description of performance	Limited and very basic description of performance	Basic and partly satisfactory description of	Good description of performance enhancing	Excellent and complete description of

© NCC Education Limited 2024

systems	enhancing technologies	enhancing technologies	performance enhancing technologies	technologies	performance enhancing technologies
Discuss the role of Data Communications and Networks in computer systems	No or very trivial discussion of data communications protocols and components	A very basic discussion of data communications protocols and components	Basic but credible discussion of data communications protocols and components	Good and wide-ranging discussion of data communications protocols and components	Excellent and complete discussion of data communications protocols and components

© NCC Education Limited 2024